Chapter 17: Equations of Lines (Phương trình Đường thẳng)

Chapter Overview

In coordinate geometry, a **line** can be represented algebraically by an equation. This powerful connection between algebra and geometry allows us to use algebraic methods to solve geometric problems and vice versa. Understanding different forms of line equations—general form, slope-intercept form, parametric form, and point-slope form—gives us flexibility in solving various problems.

In this chapter, we will:

- Learn about direction vectors and normal vectors of lines
- Write equations of lines in general form and parametric form
- Convert between different forms of line equations
- Find the angle between two lines and determine their relative positions
- Calculate the distance from a point to a line
- Apply line equations to solve real-world problems

17.1 Bilingual Vocabulary (Từ vựng Song ngữ)

Core Concepts

English	IPA	Vietnamese	Example
Line	/laɪn/	Đường thẳng	A line passes through two points
Equation of a line	/ɪˈkweɪʒən əv ə laɪn/	Phương trình đường thẳng	$ax + by + c = 0$
Direction vector	/dəˈrekʃən ˈvektər/	Vectơ chỉ phương	A vector parallel to the line
Normal vector	/ˈnɔːrməl ˈvektər/	Vectơ pháp tuyến	A vector perpendicular to the line

Slope	/sloʊp/	Hệ số góc	The slope of $y = 2x + 3$ is 2
Y-intercept	/waɪ ˈɪntərsept/	Tung độ gốc	Where the line crosses the y-axis
X-intercept	/eks ˈɪntərsept/	Hoành độ gốc	Where the line crosses the x-axis
Parameter	/pəˈræmɪtər/	Tham số	t in parametric equations

Forms of Line Equations

English	IPA	Vietnamese	Form
General form	/ˈdʒenərəl fɔːrm/	Dạng tổng quát	$ax + by + c = 0$
Slope-intercept form	/sloʊp ˈɪntərsept fɔːrm/	Dạng hệ số góc-tung độ gốc	$y = mx + b$
Point-slope form	/pɔɪnt sloʊp fɔːrm/	Dạng điểm-hệ số góc	$y - y_0 = m(x - x_0)$
Parametric form	/ˌpærəˈmetrɪk fɔːrm/	Dạng tham số	$\begin{cases} x = x_0 + at \\ y = y_0 + bt \end{cases}$
Intercept form	/ˈɪntərsept fɔːrm/	Dạng đoạn chắn	$\frac{x}{a} + \frac{y}{b} = 1$

Relative Positions

English	IPA	Vietnamese	Condition
Parallel lines	/ˈpærəlel laɪnz/	Hai đường thẳng song song	Same slope or proportional coefficients

Perpendicular lines	/ˌpɜːrpənˈdɪkjələr laɪnz/	Hai đường thẳng vuông góc	Product of slopes = -1
Intersecting lines	/ˌɪntərˈsektɪŋ laɪnz/	Hai đường thẳng cắt nhau	Lines meet at one point
Coincident lines	/koʊˈɪnsɪdənt laɪnz/	Hai đường thẳng trùng nhau	Same line, different equations

17.2 Direction Vectors and Normal Vectors (Vectơ Chỉ phương và Vectơ Pháp tuyến)

Direction Vector

> **Definition: Direction Vector**
>
> A non-zero vector u is called a **direction vector** of line ℓ if u is parallel to ℓ.
>
> **Properties:**
>
> - If u is a direction vector of ℓ, then ku ($keq0$) is also a direction vector of ℓ

> - A line is completely determined by one point and one direction vector

Normal Vector

> **Definition: Normal Vector**
>
> A non-zero vector
> n
> is called a **normal vector** of line
> ℓ
> if
> n
> is perpendicular to
> ℓ
> .
>
> **Properties:**
>
> - If
> n
>   is a normal vector of
> ℓ
>   , then
> kn
>   (
> $keq0$
>   ) is also a normal vector of
> ℓ
> - A line is completely determined by one point and one normal vector

Relationship Between Direction and Normal Vectors

> **Key Relationship**
>
> If
> $n(a, b)$
> is a normal vector of line
> ℓ
> , then

> $u(-b, a)$
> or
> $u(b, -a)$
> is a direction vector of
> ℓ
> , and vice versa.
>
> This is because
> $n \cdot u = a(-b) + b(a) = 0$

Example 17.1: Finding Direction and Normal Vectors

In the coordinate plane, triangle ABC has vertices A(3, 1), B(4, 0), C(5, 3). Find:

(a) A normal vector of the perpendicular bisector of AB

(b) A normal vector of the altitude from A

Solution:

(a) The perpendicular bisector of AB is perpendicular to AB.

$$AB = (4 - 3, 0 - 1) = (1, -1)$$

Since the perpendicular bisector is perpendicular to AB,
AB
is a normal vector.

Answer:
$n = (1, -1)$

(b) The altitude from A is perpendicular to BC.

$$BC = (5 - 4, 3 - 0) = (1, 3)$$

Since the altitude is perpendicular to BC,
BC
is a normal vector of the altitude.

Answer:
$n = (1, 3)$

17.3 General Form of Line Equations (Phương trình Tổng quát)

Derivation

Consider a line
ℓ
passing through point
$A(x_0, y_0)$
with normal vector
$n(a, b)$
.

A point
$M(x, y)$
lies on
ℓ
if and only if
$AM \perp n$
, which means:

$$AM \cdot n = 0$$

$$(x - x_0, y - y_0) \cdot (a, b) = 0$$

$$a(x - x_0) + b(y - y_0) = 0$$

Expanding:

$$ax + by - ax_0 - by_0 = 0$$

Let
$c = -ax_0 - by_0$
, we get:

> **General Form of Line Equation**
>
> $$ax + by + c = 0$$

> where
> a
> and
> b
> are not both zero, and
> $n(a, b)$
> is a normal vector of the line.

Important notes:

- Every line in the plane has an equation of the form
  $ax + by + c = 0$
- Conversely, every equation
  $ax + by + c = 0$
  (with
  a, b
  not both zero) represents a line
- The coefficients
  a, b
  give a normal vector
  $n(a, b)$

Example 17.2: Writing General Form Equation

Write the general form equation of the line passing through A(2, 1) with normal vector
$n(3, 4)$
.

Solution:

Using the point-normal form:

$$3(x - 2) + 4(y - 1) = 0$$

$$3x - 6 + 4y - 4 = 0$$

$3x + 4y - 10 = 0$

Answer:
$3x + 4y - 10 = 0$

Special Cases

1. Vertical line ($b = 0$):

- Equation: $x = m$ (or $ax + c = 0$)
- Perpendicular to the x-axis
- Normal vector: $(1, 0)$

2. Horizontal line ($a = 0$):

- Equation: $y = n$ (or $by + c = 0$)
- Perpendicular to the y-axis
- Normal vector: $(0, 1)$

3. Non-vertical line ($b eq 0$):

- Can be written as:
  $y = mx + p$

- Where
  $m = -\frac{a}{b}$
  (slope) and
  $p = -\frac{c}{b}$
  (y-intercept)

Example 17.3: Connection to Linear Functions

Write the general form equation of the line passing through A(0, b) with normal vector
$n(a, -1)$
. What is the relationship to the graph of
$y = ax + b$
?

Solution:

Using the point-normal form:

$$a(x - 0) + (-1)(y - b) = 0$$

$$ax - y + b = 0$$

Rearranging:
$y = ax + b$

Answer: The line
$ax - y + b = 0$
is exactly the graph of the linear function
$y = ax + b$
.

17.4 Parametric Form of Line Equations (Phương trình Tham số)

Derivation

Consider a line ℓ passing through point $A(x_0, y_0)$ with direction vector $u(a, b)$.

A point $M(x, y)$ lies on ℓ if and only if AM is parallel to u, which means:

$AM = tu$ for some real number t

$$(x - x_0, y - y_0) = t(a, b)$$

> **Parametric Form of Line Equation**
>
> $$\begin{cases} x = x_0 + at \\ y = y_0 + bt \end{cases}$$
>
> where t is a parameter, and $u(a, b)$ is a direction vector of the line.

Geometric interpretation: As t varies over all real numbers, the point (x, y) traces out the entire line.

Example 17.4: Writing Parametric Equations

Write the parametric equation of the line passing through A(2, -3) with direction vector $u(4, -1)$.

Solution:

Using the parametric form:

$$\begin{cases} x = 2 + 4t \\ y = -3 - t \end{cases}$$

Answer:

$$\begin{cases} x = 2 + 4t \\ y = -3 - t \end{cases}$$

Example 17.5: Line Through Two Points

Write the parametric and general form equations of the line passing through A(2, 3) and B(1, 5).

Solution:

Parametric form:

The line passes through A(2, 3) with direction vector:

$$AB = (1 - 2, 5 - 3) = (-1, 2)$$

Parametric equation:

$$\begin{cases} x = 2 - t \\ y = 3 + 2t \end{cases}$$

General form:

A normal vector is perpendicular to

$AB(-1, 2)$

, so we can use

$n(2, 1)$

.

Using point A(2, 3):

$$2(x - 2) + 1(y - 3) = 0$$

$$2x - 4 + y - 3 = 0$$

$$2x + y - 7 = 0$$

Answer:

- Parametric:
  $$\begin{cases} x = 2 - t \\ y = 3 + 2t \end{cases}$$
- General:
  $$2x + y - 7 = 0$$

17.5 Converting Between Forms (Chuyển đổi giữa các Dạng)

From General to Parametric

Given:

$$ax + by + c = 0$$

1. Find a direction vector:
   $u(-b, a)$

or
$u(b, -a)$

2. Find a point on the line (set
   $x = 0$
   or
   $y = 0$
)
3. Write parametric form using the point and direction vector

From Parametric to General

Given:

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \end{cases}$$

1. Direction vector is
   $u(a, b)$

2. Normal vector is
   $n(-b, a)$
   or
   $n(b, -a)$

3. Use point
   (x_0, y_0)
   and normal vector to write general form

Example 17.6: Converting Forms

Convert the following equations:

(a) From general to parametric:
$2x - y + 1 = 0$

(b) From parametric to general:

$$\begin{cases} x = 1 + 2t \\ y = 3 + 5t \end{cases}$$

Solution:

(a) From

$2x - y + 1 = 0$

:

- Normal vector:

  $n(2, -1)$

- Direction vector:

  $u(1, 2)$

  (perpendicular to normal)

- Find a point: Let

  $x = 0$

  , then

  $-y + 1 = 0$

  , so

  $y = 1$

  . Point:

  $(0, 1)$

Parametric form:

$\{ \begin{matrix} x = 0 + t = t \\ y = 1 + 2t \end{matrix}$

or

$\{ \begin{matrix} x = t \\ y = 1 + 2t \end{matrix}$

(b) From

$\{ \begin{matrix} x = 1 + 2t \\ y = 3 + 5t \end{matrix}$

:

- Direction vector:

  $u(2, 5)$

- Normal vector:

  $n(-5, 2)$

or

$n(5, -2)$

- Point:

  $(1, 3)$

Using

$n(5, -2)$

:

$$5(x - 1) - 2(y - 3) = 0$$

$$5x - 5 - 2y + 6 = 0$$

$$5x - 2y + 1 = 0$$

Answer:

- (a)

  $\begin{cases} x = t \\ y = 1 + 2t \end{cases}$

- (b)

  $5x - 2y + 1 = 0$

17.6 Relative Positions of Two Lines (Vị trí Tương đối của Hai đường thẳng)

Consider two lines:

- $\ell_1 : a_1x + b_1y + c_1 = 0$
- $\ell_2 : a_2x + b_2y + c_2 = 0$

> **Relative Positions**

> 1. **Parallel** (
> $\ell_1 \parallel \ell_2$
>   ):
> $\frac{a_1}{a_2} = \frac{b_1}{b_2} eq \frac{c_1}{c_2}$
>
> 2. **Coincident** (
> $\ell_1 \equiv \ell_2$
>   ):
> $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
>
> 3. **Intersecting** (
> ℓ_1
>   intersects
> ℓ_2
>   ):
> $\frac{a_1}{a_2} eq \frac{b_1}{b_2}$
>
> 4. **Perpendicular** (
> $\ell_1 \perp \ell_2$
>   ):
> $a_1 a_2 + b_1 b_2 = 0$

Note: The perpendicular condition comes from the dot product of normal vectors: $n_1 \cdot n_2 = 0$

Example 17.7: Determining Relative Positions

Determine the relative position of each pair of lines:

(a)
$\ell_1 : 2x - 3y + 5 = 0$
and
$\ell_2 : 4x - 6y + 7 = 0$

(b)
$\ell_1 : 3x + 4y - 1 = 0$
and
$\ell_2 : 4x - 3y + 2 = 0$

Solution:

(a) Check ratios:
$\frac{a_1}{a_2} = \frac{2}{4} = \frac{1}{2}$
,
$\frac{b_1}{b_2} = \frac{-3}{-6} = \frac{1}{2}$
,
$\frac{c_1}{c_2} = \frac{5}{7}$

Since
$\frac{a_1}{a_2} = \frac{b_1}{b_2} eq \frac{c_1}{c_2}$
, the lines are **parallel**.

(b) Check perpendicularity:

$$a_1 a_2 + b_1 b_2 = (3)(4) + (4)(-3) = 12 - 12 = 0$$

The lines are **perpendicular**.

17.7 Angle Between Two Lines (Góc giữa Hai đường thẳng)

The angle
θ
between two lines
ℓ_1
and
ℓ_2
(
$0° \leq \theta \leq 90°$
) can be found using their normal vectors.

> **Angle Between Two Lines**

> If
> $\ell_1 : a_1 x + b_1 y + c_1 = 0$
> and
> $\ell_2 : a_2 x + b_2 y + c_2 = 0$
> , then:
> $\cos\theta = \frac{|a_1 a_2 + b_1 b_2|}{\sqrt{a_1^2 + b_1^2} \cdot \sqrt{a_2^2 + b_2^2}}$

Example 17.8: Finding Angle Between Lines

Find the angle between the lines
$\ell_1 : x + y - 1 = 0$
and
$\ell_2 : x - \sqrt{3}y + 2 = 0$
.

Solution:

Normal vectors:
$n_1(1, 1)$
and
$n_2(1, -\sqrt{3})$

$$\cos\theta = \frac{|1 \cdot 1 + 1 \cdot (-\sqrt{3})|}{\sqrt{1^2 + 1^2} \cdot \sqrt{1^2 + (-\sqrt{3})^2}}$$

$$= \frac{|1 - \sqrt{3}|}{\sqrt{2} \cdot \sqrt{4}} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\theta = \arccos\left(\frac{\sqrt{6} - \sqrt{2}}{4}\right) = 75°$$

Answer: The angle between the lines is 75°.

17.8 Distance from a Point to a Line (Khoảng cách từ Điểm đến Đường thẳng)

> **Distance Formula**
>
> The distance from point $M_0(x_0, y_0)$ to line $\ell : ax + by + c = 0$ is:
>
> $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$

Derivation: This formula comes from projecting the vector from any point on the line to M_0 onto the normal vector.

Example 17.9: Distance from Point to Line

Find the distance from point A(1, 2) to the line $\ell : 3x - 4y + 5 = 0$.

Solution:

Using the distance formula:

$d = \frac{|3(1) - 4(2) + 5|}{\sqrt{3^2 + (-4)^2}}$

$= \frac{|3 - 8 + 5|}{\sqrt{9 + 16}} = \frac{|0|}{\sqrt{25}} = \frac{0}{5} = 0$

Answer: The distance is 0, which means point A lies on the line.

Example 17.10: Distance Between Parallel Lines

Find the distance between the parallel lines $\ell_1 : 3x + 4y - 5 = 0$

and
$\ell_2 : 3x + 4y + 10 = 0$
.

Solution:

Since the lines are parallel, we can find the distance by:

1. Choosing any point on
   ℓ_1
2. Finding its distance to
   ℓ_2

Let's find a point on
ℓ_1
. Set
$x = 0$
:
$4y - 5 = 0 \Rightarrow y = \frac{5}{4}$

Point:
$P(0, \frac{5}{4})$

Distance from P to
ℓ_2
:
$d = \frac{|3(0)+4(\frac{5}{4})+10|}{\sqrt{3^2+4^2}}$

$= \frac{|0+5+10|}{\sqrt{25}} = \frac{15}{5} = 3$

Answer: The distance between the parallel lines is 3.

17.9 Applications (Ứng dụng)

Application 1: Temperature Conversion

Problem: The conversion between Celsius (°C) and Fahrenheit (°F) is determined by two reference points:

- Water freezes at 0°C = 32°F
- Water boils at 100°C = 212°F

If we plot points (C, F) on a coordinate plane, they lie on a line through A(0, 32) and B(100, 212).

(a) Find the equation of this line.

(b) What is 0°F in Celsius? What is 100°F in Celsius?

Solution:

(a) Direction vector:

$AB = (100, 180)$

or simplified

$u(5, 9)$

Using point A(0, 32):

Parametric form:

$$\begin{cases} C = 5t \\ F = 32 + 9t \end{cases}$$

Eliminating

t

: From the first equation,

$t = \frac{C}{5}$

Substituting:

$F = 32 + 9 \cdot \frac{C}{5} = 32 + \frac{9C}{5}$

Or:

$F = \frac{9}{5}C + 32$

(b) For 0°F:

$$0 = \frac{9}{5}C + 32$$
$$C = -\frac{32 \cdot 5}{9} = -\frac{160}{9} \approx -17.8°C$$

For 100°F:
$$100 = \frac{9}{5}C + 32$$
$$68 = \frac{9}{5}C$$
$$C = \frac{68 \cdot 5}{9} = \frac{340}{9} \approx 37.8°C$$

Answer:

- (a)
$$F = \frac{9}{5}C + 32$$

- (b) 0°F ≈ -17.8°C; 100°F ≈ 37.8°C

Application 2: Flight Path

Problem: According to Google Maps, Noi Bai Airport (Hanoi) is at 21.2°N latitude, 105.8°E longitude. Da Nang Airport is at 16.1°N latitude, 108.2°E longitude. A plane flies from Noi Bai to Da Nang. At time
t
hours after departure, the plane's position (latitude x°N, longitude y°E) is:

$$\begin{cases} x = 21.2 - \frac{51}{10}t \\ y = 105.8 + \frac{24}{10}t \end{cases}$$

(a) How long does the flight take?

(b) At
$t = 1$
hour, has the plane crossed the 17°N parallel?

Solution:

(a) The plane reaches Da Nang when
$x = 16.1$
:

$$16.1 = 21.2 - \frac{51}{10}t$$

$\frac{51}{10}t = 5.1$

$t = \frac{5.1 \cdot 10}{51} = 1$
hour

Answer: The flight takes 1 hour.

(b) At
$t = 1$
:

$x = 21.2 - \frac{51}{10}(1) = 21.2 - 5.1 = 16.1°N$

Since 16.1° < 17°, yes, the plane has crossed the 17°N parallel.

Answer: Yes, at
$t = 1$
hour, the plane is at 16.1°N, which is south of 17°N.

17.10 Practice Exercises (Bài tập Luyện tập)

Exercise 17.1: Write the general form equation of the line passing through A(1, 3) with normal vector
$n(2, 1)$
.

Exercise 17.2: Write the parametric equation of the line passing through B(-2, 1) with direction vector
$u(3, 2)$
.

Exercise 17.3: Write both parametric and general form equations of the line passing through A(1, 2) and B(3, 0).

Exercise 17.4: Determine the relative position of the lines:

- $\ell_1 : 2x + 3y - 5 = 0$
- $\ell_2 : 4x + 6y - 10 = 0$

Exercise 17.5: Find the distance from point M(2, -1) to the line
$3x + 4y - 12 = 0$
.

17.11 Chapter Summary (Tóm tắt Chương)

Key Concepts

1. **Direction and Normal Vectors:**
   - Direction vector
     $u(a, b)$
     is parallel to the line
   - Normal vector
     $n(a, b)$
     is perpendicular to the line
   - If
     $n(a, b)$
     is normal, then
     $u(-b, a)$
     is direction
2. **Forms of Line Equations:**
   - **General:**
     $ax + by + c = 0$
   - **Parametric:**
     $\{\begin{matrix} x = x_0 + at \\ y = y_0 + bt \end{matrix}$
   - **Slope-intercept:**
     $y = mx + b$
3. **Relative Positions:**
   - Parallel:
     $\frac{a_1}{a_2} = \frac{b_1}{b_2} eq \frac{c_1}{c_2}$

- Perpendicular:
  $a_1 a_2 + b_1 b_2 = 0$

4. **Distance Formula:**
   - $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$

Important Formulas Summary

Formula	Description
$ax + by + c = 0$	General form, $n(a, b)$ is normal
$\{\begin{matrix} x = x_0 + at \\ y = y_0 + bt \end{matrix}$	Parametric form, $u(a, b)$ is direction
$a_1 a_2 + b_1 b_2 = 0$	Perpendicularity condition
$$d = \frac{	ax_0 + by_0 + c

17.12 Review Exercises (Bài tập Ôn tập)

Review 17.1: In triangle ABC with A(-1, 5), B(2, 3), C(6, 1):

- (a) Write the equation of the altitude from A
- (b) Write the equation of the median from B

Review 17.2: Find the value of m such that the line $mx + 2y - 1 = 0$ is:

- (a) Parallel to the line $3x + 6y + 5 = 0$

- (b) Perpendicular to the line
  $$2x - y + 3 = 0$$

Review 17.3: Find the distance between the parallel lines:

- $\ell_1 : 5x - 12y + 3 = 0$
- $\ell_2 : 5x - 12y - 10 = 0$

Review 17.4: A line passes through M(2, 1) and makes an angle of 45° with the positive x-axis. Write its equation.

Review 17.5: Find the coordinates of the point on the line
$$2x - y + 3 = 0$$
that is closest to the point A(1, 4).

Memory Aids (Gợi nhớ)

Direction vs. Normal vectors:

- **Direction** → **D**irection of travel (parallel to line)
- **Normal** → **N**ormal means perpendicular (⊥ to line)

Converting between vectors:

- If
  $$n(a, b)$$
  is normal, then
  $$u(-b, a)$$
  is direction
- "Swap and negate one" →
  $$(a, b) \to (-b, a)$$

Perpendicular lines:

- Dot product of normal vectors = 0
- $a_1 a_2 + b_1 b_2 = 0$

Distance formula mnemonic:

- "Plug in point, divide by length of normal"
- $d = \frac{|\text{plug in point}|}{|\text{normal vector}|}$

Common mistakes to avoid:

- Don't confuse direction and normal vectors
- Remember that a and b in $ax + by + c = 0$ give the **normal** vector, not direction
- When checking parallel lines, all three ratios must be checked
- The distance formula requires absolute value in the numerator

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End of Chapter 17